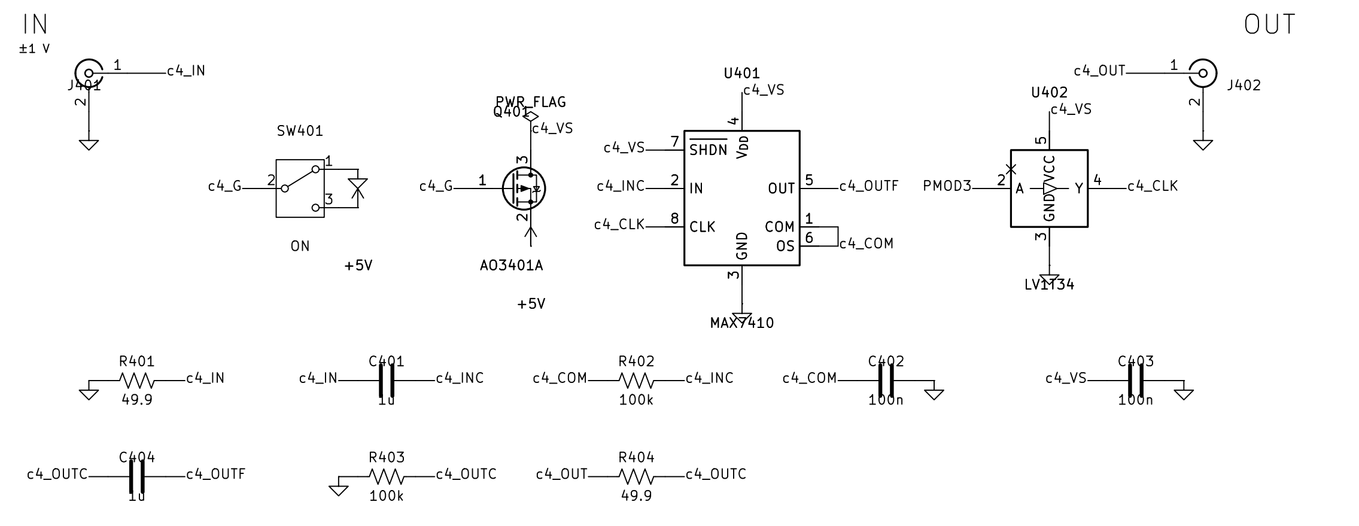
c4 SC filter, MAX7410

tutorial Switched capacitors (+5 V)

cell (0,0)



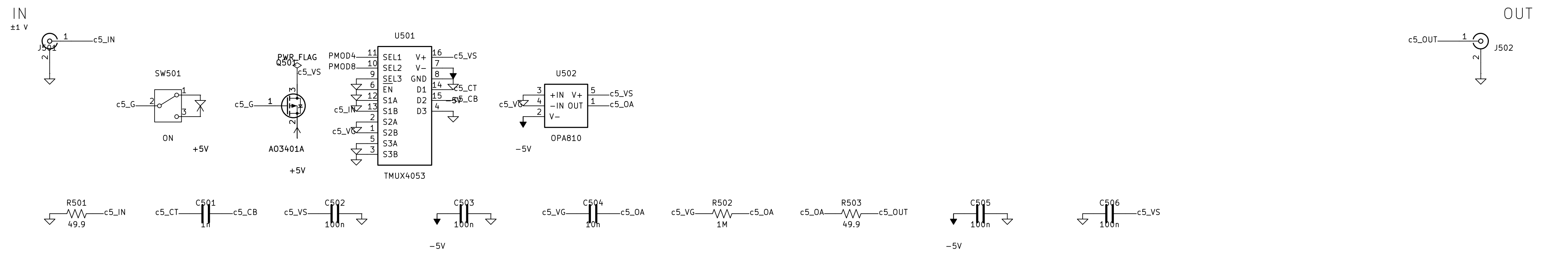
EXPECT: a 5th–order Butterworth low–pass at fCLK / 100 (up to 15 kHz from a 1.5 MHz clock on IO3); the clock's images near fCLK

MAX7410: a switched–capacitor ladder whose corner is the clock divided by 100, clocked from IO3 through an SN74LV1T34.
IN and OUT sit at COM (2.5 V), so both are AC–coupled. Sweep the clock and watch the corner follow; alias a tone.

c5 SC integrator, TMUX4053

tutorial Switched capacitors (+–5 V)

cells (1,0), (2,0)



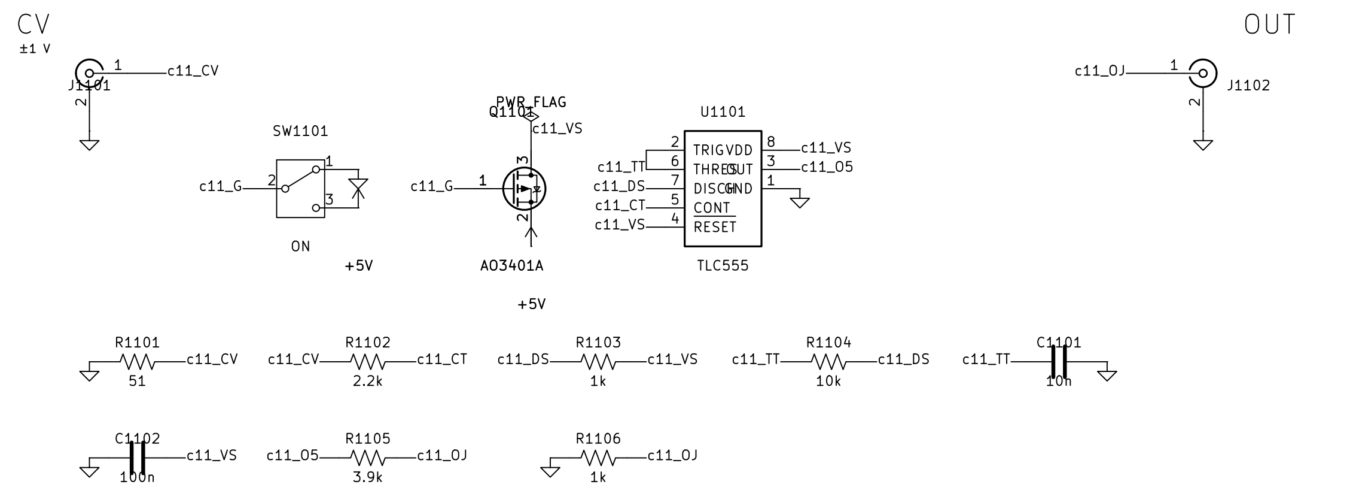
EXPECT: an integrator whose input resistor is 1 / (f x 1 nF): 1 M at 1 kHz, 1 k at 1 MHz; OUT = –(1 / R C) x the integral of IN

phi1 (IO4) connects the 1 nF's top plate to IN, phi2 (IO8) its bottom plate to the summing node: C x Vin per cycle at f,
a resistor of 1 / (f C). Make the phases non–overlapping, or overlap them and watch IN leak through. 1 M sets the DC gain.

c11 555, TLC555

tutorial Trading speed for bits (+5 V)

cell (3,0)



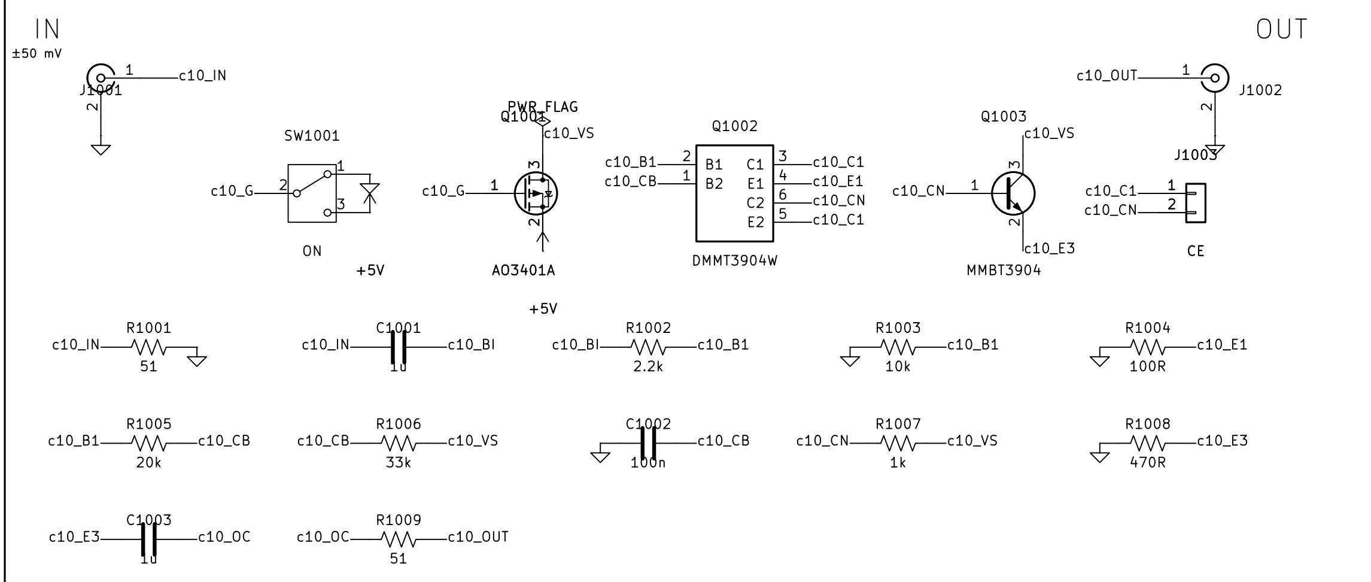
EXPECT: a square near 6.9 kHz on OUT (5 V divided to about 1 V); CV moves it: up as CV falls, down as it rises

TLC555 astable: f = 1.44 / ((1 k + 2 x 10 k) x 10 nF), the capacitor between 1/3 and 2/3 of the supply. CV pulls the
upper threshold through 2.2 k: a voltage–to–frequency converter. Count OUT's cycles in the FPGA and you have an ADC.

c10 cascode

tutorial Transistors (+5 V)

cell (0,1)



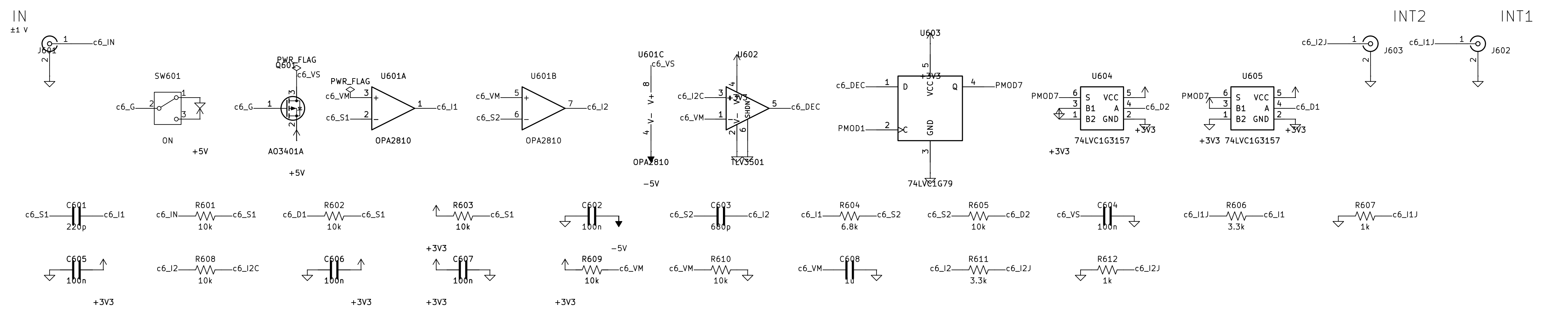
EXPECT: gain about –6 into a high impedance (–8 at Q1's collector), half that into 50 ohm; common emitter (jumper on) about 2 MHz wide, the cascode (jumper parked) several times wider

Q1's collector–base capacitance, multiplied by the gain (Miller), meets the 2.2 k at its base; the cascode holds Q1's
collector still, so only one Ccb counts. The jumper on the header shorts the cascode out: compare the two sweeps.

c6 delta–sigma, second order

tutorial Trading speed for bits (+–5 V)

cells (1,1), (2,1)



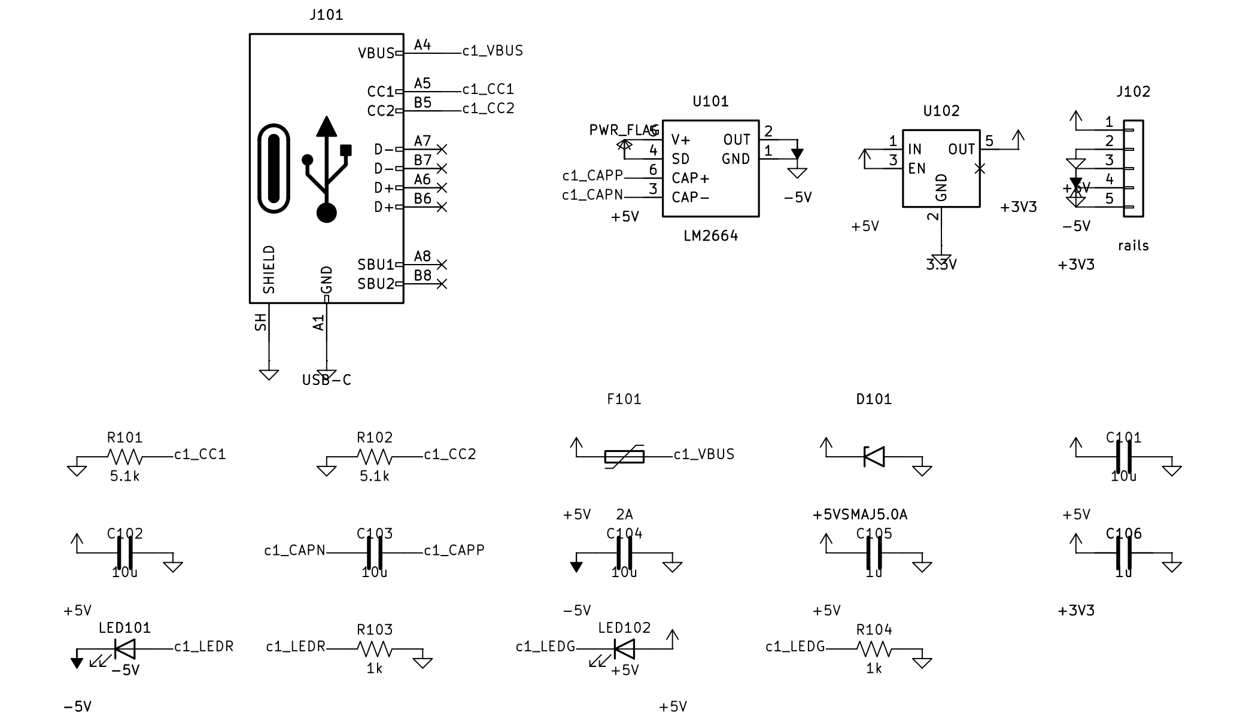
EXPECT: a one–bit stream on IO7 whose density follows IN (+–1.65 V full scale), its noise shaped at 40 dB per decade

Clock IO1 at about 1 MHz and count IO7's ones. The second integrator pushes the quantization noise up at 40 dB per
decade (AMOD–A's a6 is first order: 20 dB), so a sinc3 decimator gets more bits per clock. INT1, INT2: high impedance.

c1 power, USB–C 5 V in: +5, –5, +3.3 out

tutorial Regulators, Switched capacitors

cell (3,1)



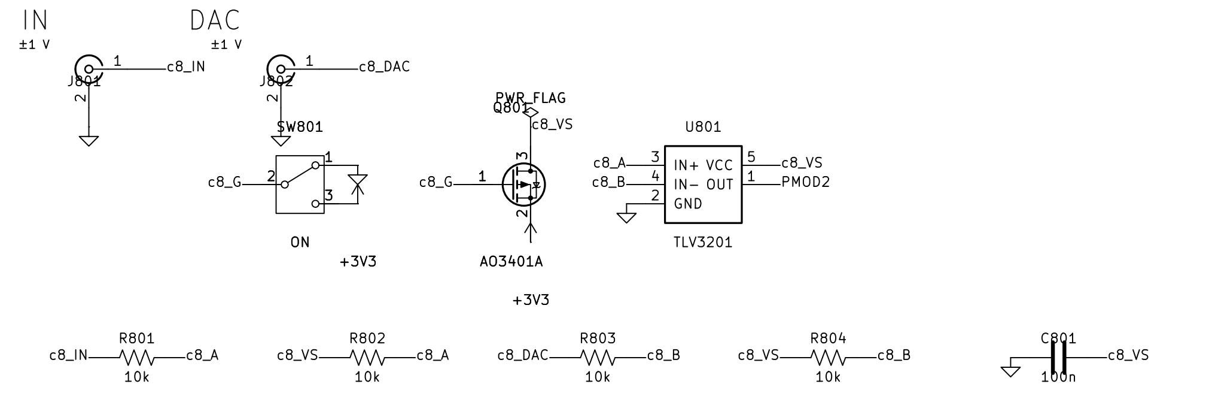
EXPECT: +5 V from the USB–C (up to 1.5 A), –5 V at 40 mA, +3.3 V at 300 mA; the green LED says 5 V is in, the red one that –5 V is up

USB–C power only: 5.1 k on each CC asks any source for 5 V. A 2 A PTC, a 5 V TVS (clamps at 9.2 V) and 20 uF.
The LM2664 charge pump makes –5 V for the op–amp tiles (40 mA: four tiles at 10 mA); it is not switched per tile, an op
amp with its +5 V removed draws nothing. The TPS7A2033 makes 3.3 V for the comparator and the lines that meet the FPGA.
The socket along the top carries +5, ground, –5, ground, +3.3: the bench's rails for a breadboard experiment.

c8 comparator, TLV3201

tutorial Trading speed for bits (+3.3 V)

cell (0,2)



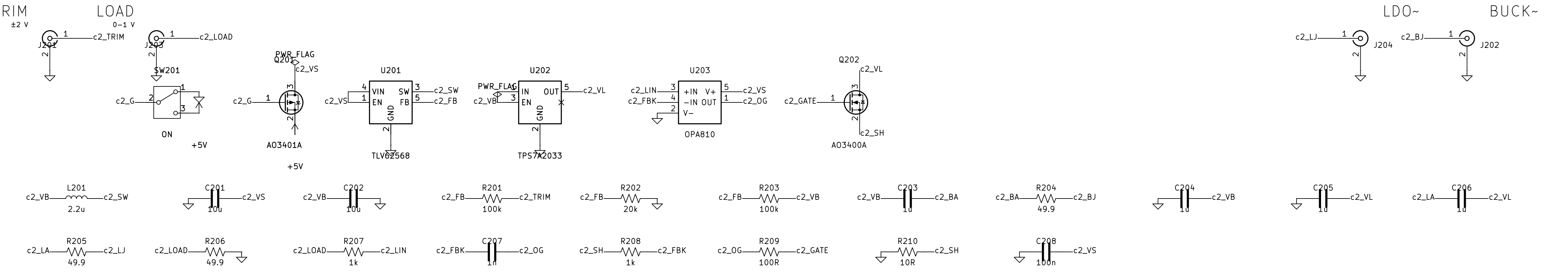
EXPECT: BIT on Pmod IO2 is high while IN is above DAC; a SAR search in gateware gives 12 bits in 12 steps of the DAC

TLV3201 (40 ns, 1.2 mV hysteresis) on the 3.3 V rail. Each input is averaged with the rail through a matched pair
of 0.1 % 10 k, so +–1 V lands at 1.15–2.15 V on both pins and the comparison is unchanged. 20 k input impedance:
drive it from a 50 ohm source, read the bit on IO2. A successive–approximation ADC: the gateware sets the host's
DAC, reads the bit, halves the step; a tracking ADC: up when high, down when low; a slope ADC with a ramp on DAC.

c2 regulators, buck, LDO, load

tutorial Regulators (+5 V)

cells (1,2), (2,2)



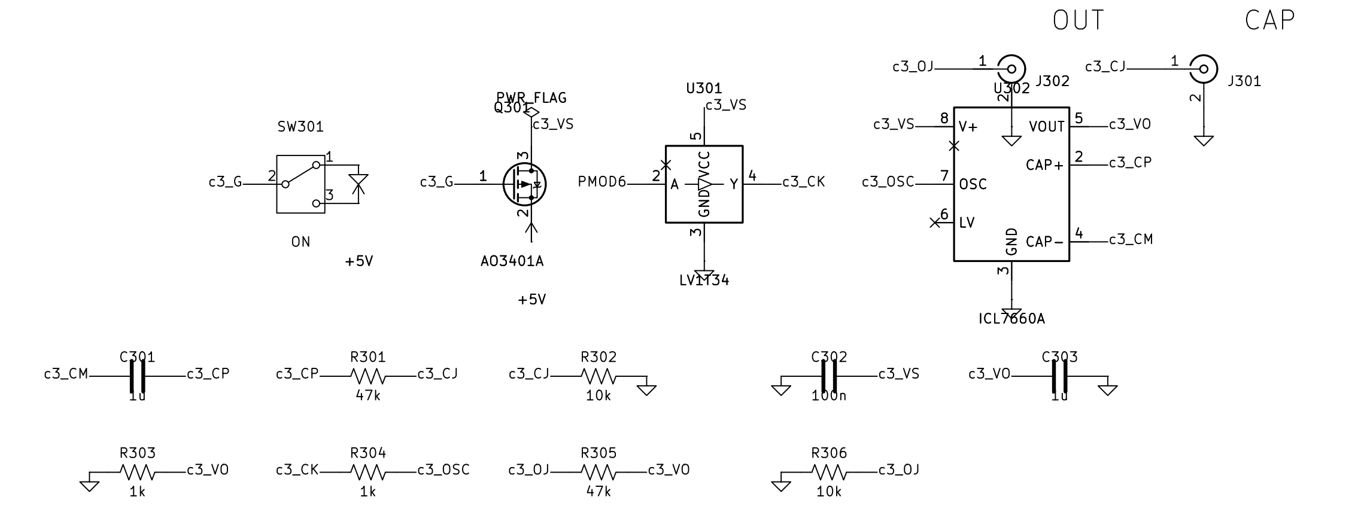
EXPECT: BUCK~ shows the buck's 1.5 MHz ripple (a few mV, pulse–skipping below about 150 mA); LDO~ the same after the LDO, 40 dB smaller; LOAD sets 100 mA per volt

TLV62568: 3.6 V from 100 k / 20 k; TRIM through 100 k into the feedback node moves it (0 V: 4.2 V, +1 V: 3.2 V, open: 3.6 V).
TPS7A2033: 3.3 V, about 92 dB of PSRR at 1 kHz and 40 dB at 1 MHz; below 3.5 V in it drops out (lower TRIM's voltage).
The load: an OPA810 servos 10 ohm to LOAD's voltage through an AO3400A (1 k and 1 nF slow its loop): a square on LOAD
is a load step. Keep it under 100 mA for long (the shunt is 1/8 W), steps can be larger. Both outputs are AC–coupled.

c3 charge pump, ICL7660A

tutorial Regulators (+5 V)

cell (3,2)



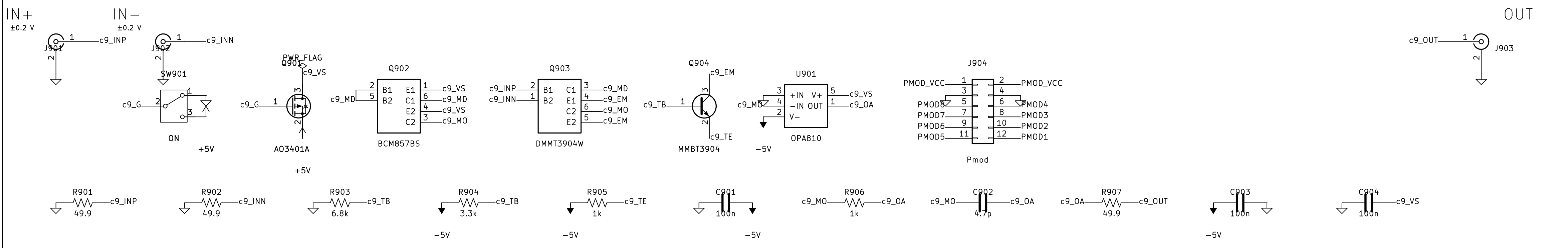
EXPECT: about –5 V on the output, falling with load as 1 / (f C) + the switches: 2 k at 1 kHz, 80 ohm at 100 kHz (the pump at half IO6's clock)

ICL7660A: switches charge the flying 1 uF across the supply and dump it, upside down, into the output's 1 uF. IO6
clocks it through an SN74LV1T34 and 1 k; with 1 k of load, sweep the clock and read OUT: Rout = 1 / (f C) + 60 ohm.

c9 differential pair

tutorial Transistors (+–5 V)

cells (0,3), (1,3)



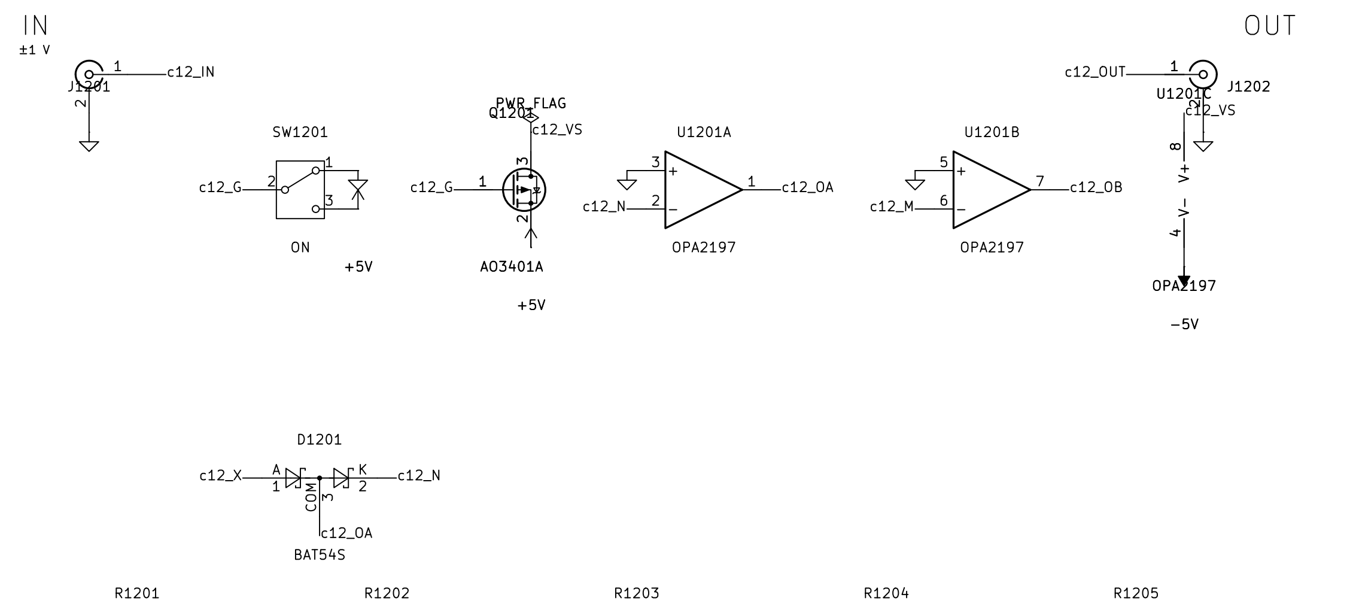
EXPECT: OUT = –1 k x (Ic1 – Ic2): +–1 V, tanh of (IN+ – IN–) / 2 VT, linear within about +–25 mV of balance

A matched NPN pair (DMMT3904W: 2 mV of VBE, 2 % of hFE) on a 1 mA tail; a matched PNP pair as the mirror; an OPA810
holds the output node at 0 V and turns the difference current into volts. Sweep IN+ with IN– at 0 V: the tanh, its
slope gm = Itail / (4 VT), about 10 mS here; a triangle in becomes a sine–like wave out when it is driven past balance.

c12 rectifier, precision

tutorial Transistors (+–5 V)

cell (2,3)



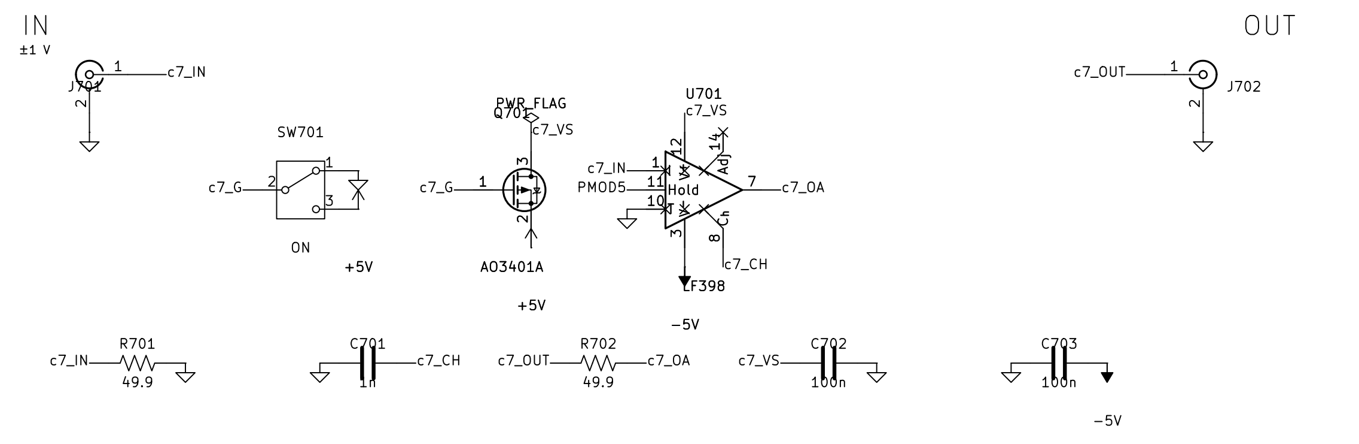
EXPECT: OUT = |IN| from DC to about 100 kHz; above that the first stage's slew across the diodes leaves a glitch at each zero crossing

Stage 1: inverting with BAT54S in its feedback, so its node X follows –IN for positive IN and stays at 0 for negative.
Stage 2: OUT = –(IN + 2 X) = |IN|. No dead zone from the diodes (the op amp's gain hides their drop) until the op amp
must slew through two diode drops at each zero crossing: watch the glitch grow with frequency.

c7 track and hold, LF398

tutorial Trading speed for bits (+–5 V)

cell (3,3)



EXPECT: OUT follows IN while IO5 is high and holds when it falls: 4 us to acquire, about 30 mV/s of droop, a few mV of hold step

LF398 on +–5 V (its minimum supply: inputs within +–1.5 V). 1 nF C0G: acquisition about 4 us, droop about 30 mV/s.
Measure the acquisition time, the droop, the pedestal (the hold step) against IN, the feedthrough of a sine while
holding, and the aperture delay against the FPGA's edge on IO5.

no mounting holes: four stick–on rubber feet, marked on the back

ground loops for the scope's clips, one per channel, in the board's margins

Drawn by scripts/gen_sch.py from boards/amod_c.py; the expected responses from sim/passive.py
Every tile is one circuit with its own SMA jacks; the board's silkscreen draws each tile's wires as they
a buck and an LDO under load, a charge pump, switched capacitors, a second–order delta–sigma, sa
Board: Jason Gallicchio and Claude | open hardware, CERN–OHL–P–2.0 | github.com/gallicchio/
Sheet: /
File: amod_c.kicad_sch
Title: AMOD–C: power and data conversion
Size: A1
Date: 2026–10–25
Rev: A
KiCad E.D.A. 10.0.7
Id: 1/1